Lecture 13: Two-Stage Least Squares (2SLS)

Last lecture, we introduced instrumental variables (IV), focusing on the case with a single endogenous regressor and a single instrument. Today, we will talk about how to implement instrumental variables in a setting with multiple regressors and/or instruments. We will do this using the method called two-stage least squares (abbreviated 2SLS or TSLS). This will also allow us to discuss the case of overidentification: a setting where we have more instruments than endogenous regressors.

Refresher on Instrumental Variables: RCT with Imperfect Compliance

To begin, let's refresh ourselves on instrumental variables. Suppose that we are interested in how attending a charter school affects test scores. Our sample will be all of the charter school's applicants for a given cohort. We are interested in estimating the following linear model:

$$\text{Test Score}_i = \alpha + \beta \cdot \text{Attended Charter}_i + \varepsilon_i$$

where Test Score_i is student i's score on a standardized exam, and $\text{Attended Charter}_i$ is a dummy variable equal to one if the student attends the charter school. Rather than estimating this model with OLS, we instead will use instrumental variables. The charter school runs a lottery among all applicants, and makes initial offers to the lottery winners. The dummy variable Lottery Winner_i is equal to one if student i is one of the lottery winners.

However, not everyone who receives an offer from the charter school will actually attend. Moreover, after the initial lottery, the charter will admit some students (non-randomly) off of the waitlist. So, although Lottery Winner_i will be strongly positively correlated with $\text{Attended Charter}_i$, they will not be the same.

This setup is equivalent to what we call a randomized controlled trial (RCT) with imperfect compliance: there is an initial assignment to treatment and control, but the actual assignment may differ from the assignment. In this setting, we can still estimate valid treatment effects by using the initial assignment as an instrument. That is what we will do here: we will use Lottery Winner_i as an instrument to estimate the effect of charter attendance.

Exercise 1. What is the orthogonality condition for OLS to be consistent?

Exercise 2. Why might the OLS orthogonality condition be violated? What might cause charter attendance to be endogenous?

Exercise 3. What is the orthogonality condition for IV?

Exercise 4. What does it mean for the instrument to be as-good-as-randomly assigned? What does it mean for the instrument to satisfy the exclusion restriction?

Exercise 5. What is the first stage regression?

Exercise 6. What would the reduced form regression be? How can we combine this with the first stage to get our IV estimate?

Suppose now that there are actually two charter schools, Charter 1 and Charter 2, and we wish to estimate separate effects of each charter. We wish to run the regression:

$$\text{Test Score}_i = \alpha + \beta_1 \cdot \text{Attended Charter 1}_i + \beta_2 \cdot \text{Attended Charter 2}_i + \varepsilon_i$$

where $\text{Attended Charter 1}_i$ is a dummy for attending Charter 1, and $\text{Attended Charter 2}_i$ is a dummy for attending Charter 2. We will again use the charter lotteries as instruments, but this time we want to use two instruments: $\text{Lottery 1 Winner}_i$ is a dummy equal to one if the student was selected in Charter 1's lottery, and $\text{Lottery 2 Winner}_i$ is equal to one if the student was selected in Charter 2's lottery (suppose for simplicity that everyone in our sample applied to both lotteries). Today, we will study how to implement an IV regression like this, with multiple regressors and instruments.

Two-Stage Least Squares

So far, we have seen how to do instrumental variables for a single regressor and a single instrument. We will now study how to run instrumental variables with multiple regressors and/or multiple instruments.

Our regression will have three different types of variables:

Endogenous Regressors: We will have K endogenous regressors, that we will denote $X_1, ..., X_K$. These are the endogenous variables that we need instruments for.

Excluded Instruments: We will have J excluded instruments, that we will denote $Z_1, ..., Z_J$. We will use these instruments to identify the coefficients on the endogenous regressors; these instruments will not be included in the regression itself.

Exogenous Regressors: We *may* also have exogenous regressors, that we will denote $W_1, ..., W_L$. These are variables for which we do not need excluded instruments. In practice, these are often additional variables that are included as controls.

We are interested in estimating the regression

$$Y_i = \alpha + \beta_1 X_{1,i} + \cdots + \beta_K X_{K,i} + \delta_1 W_{1,i} + \cdots + \delta_L W_{L,i} + \varepsilon_i$$

where Y_i is our outcome and ε_i is the error term.

To implement two-stage least squares, we will proceed in two stages:

First Stage

For each endogenous regressor, $X_{k,i}$ we run the first-stage regression:

$$X_{k,i} = \pi_0^k + \pi_1^k Z_{1,i} + \cdots + \pi_J^k Z_{J,i} + \pi_{J+1}^k W_{1,i} + \cdots + \pi_{J+L}^k W_{L,i} + u_i^k$$

where $\pi_0^k, ..., \pi_{J+L}^k$ are coefficients in the first-stage regression for $X_{k,i}$, and u_i^k is the error term of that regression. From each of these first-stage regressions, we generate predicted values for each endogenous regressor:

$$\hat{X}_{ki} = \hat{\pi}_0^k + \hat{\pi}_1^k Z_{1,i} + \cdots + \hat{\pi}_J^k Z_{J,i} + \hat{\pi}_{J+1}^k W_{1,i} + \cdots + \hat{\pi}_{J+L}^k W_{L,i}$$

Second Stage

We then run our second-stage regression using the fitted values:

$$Y_i = \alpha + \beta_1 \hat{X}_{1,i} + \cdots + \beta_K \hat{X}_{K,i} + \delta_1 W_{1,i} + \cdots + \delta_L W_{L,i} + \omega_i$$

where ω_i is the error term of this regression (note that this is not the same as the error ε_i). The estimates from this regression are our two-stage least squares estimates.

Three Notes

- The coefficients from the second-stage regression will be the correct estimates, but the standard errors from that regression are not the correct standard errors. In practice, you should implement this using Stata's built-in IV command, which will also give you the correct standard errors.
- For the exogenous regressors, we did not bother doing the first-stage prediction, and just used them directly. But actually what is going on is that the exogenous regressors are included as both regressors and instruments. Since the first-stage regression for W_1 includes W_1 on the right-hand side, it is predicted perfectly and $\hat{W}_1 = W_1$.
- You need to include the same instruments and controls in each first-stage regression, and also need to include the same controls in the second-stage regression. You can't mix and match!

Conditions for Consistency of Two-Stage Least Squares

For two-stage least squares to be consistent, we need the following to be true:

Assumption (IID Sampling)**.** *Each observation is drawn independently and from the identical distribution as the other observations.*

Assumption (Full Rank Condition (Relevance + No Perfect Collinearity))**.** *None of the predicted regressors in the second stage ($\hat{X}_{k,i}$ or $W_{l,i}$) is perfectly collinear with the other predicted regressors.*

Assumption (Orthogonality Condition)**.** *The error term, ε_i, has zero mean $\mathbb{E}[\varepsilon_i] = 0$, and is uncorrelated with each instrument, $Cov(Z_{j,i}, \varepsilon_i) = 0$ for all j, and with each exogenous regressor $Cov(W_{l,i}, \varepsilon_i) = 0$ for all l.*

These are the same assumptions as we had for the single-regressor, single-instrument case, but extended to the general setting. IID sampling is the same as before. The "full rank condition" is a generalization of the relevance condition: note that if the instrument had no correlation with the regressor (failure of the relevance condition), then it would produce no variation in the predicted values of the regressor, and the full rank condition would fail. Finally, our orthogonality condition is the same as before, but now extended to cover all of the instruments, as well as the exogenous regressors (which are in fact instruments themselves).

Under our three assumptions for instrumental variables, two-stage least squares is consistent:

Theorem (Two-Stage Least Squares is Consistent)**.** *Suppose that each observation is sampled iid, the full rank condition is satisfied, and the orthogonality condition holds. Then, two-stage least squares is consistent: $\hat{\beta}_k^{2SLS} \xrightarrow{p} \beta_k$ for all k, and $\hat{\delta}_l^{2SLS} \xrightarrow{p} \delta_l$ for all l.*

How Many Instruments Do We Need?

It turns out that we need to have at least as many excluded instruments as we have endogenous regressors: we must have $J \geq K$. We can understand why this is by examining what happens in three cases.

Just Identified The first case is when we have the same number of excluded instruments as we have endogenous regressors, $K = J$. We call this the just-identified case. In this case, 2SLS will set the sample analogue of each of the orthogonality conditions to be exactly equal to zero. That is, 2SLS will select coefficients to exactly solve the system of linear equations:

$$\begin{aligned} \bar{\varepsilon} &= 0 \\ \widehat{\text{Cov}}(Z_{j,i}, \varepsilon_i) &= 0 \text{ for all } j \\ \widehat{\text{Cov}}(W_{l,i}, \varepsilon_i) &= 0 \text{ for all } l \end{aligned}$$

This will give us exactly as many equations as we have unknown parameters, and so solving the equations will pin down the parameters.

Overidentified The second case is when we have more excluded instruments than endogenous regressors, $J > K$. We call this the overidentified case. In this case, 2SLS will generally not be able to set $\widehat{\text{Cov}}(Z_{j,i}, \varepsilon_i) = 0$ for all j. Instead, it will select parameters that set these covariances as close to zero as it can.

Underidentified The final case is when we have fewer excluded instruments than endogenous regressors, $J < K$. We call this the underidentified case. In this case, we do not have enough equations to pin down all the unknown parameters. If we try to run 2SLS, our software will throw an error. In particular, the full rank condition will fail: if we have fewer instruments than endogenous regressors, then there will not be enough variation in the predicted regressors $\hat{X}_{k,i}$, and some of them will be perfectly collinear with the others.

Equivalence of Simple IV and 2SLS

Last lecture, we studied the case of a single regressor and a single instrument, and we derived the formula:

$$\hat{\beta}^{IV} = \frac{\widehat{\text{Cov}}(Y_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)}$$

We can show that if we applied 2SLS in this setting, it is equivalent: it will simplify to exactly the same estimator. To see this, note that our first stage is:

$$X_i = \gamma + \pi \cdot Z_i + u_i$$

Estimating this yields the predicted regressor $\hat{X}_i = \hat{\gamma} + \hat{\pi} \cdot Z_i$, where $\hat{\pi} = \frac{\widehat{\text{Cov}}(X_i, Z_i)}{\widehat{\text{Var}}(Z_i)}$. We can then run the second stage:

$$Y_i = \alpha + \beta \cdot \hat{X}_i + \omega_i$$

Which yields

$$\begin{aligned}
\hat{\beta}^{2SLS} &= \frac{\widehat{\text{Cov}}\left(\hat{X}_i, Y_i\right)}{\widehat{\text{Var}}\left(\hat{X}_i\right)} = \frac{\widehat{\text{Cov}}\left(\hat{\gamma} + \hat{\pi} \cdot Z_i, Y_i\right)}{\widehat{\text{Var}}\left(\hat{\gamma} + \hat{\pi} \cdot Z_i\right)} \\
&= \frac{\hat{\pi} \cdot \widehat{\text{Cov}}(Z_i, Y_i)}{\hat{\pi}^2 \cdot \widehat{\text{Var}}(Z_i)} = \frac{\widehat{\text{Cov}}(Z_i, Y_i)}{\hat{\pi} \cdot \widehat{\text{Var}}(Z_i)} \\
&= \frac{\widehat{\text{Cov}}(Z_i, Y_i)}{\left(\widehat{\text{Cov}}(X_i, Z_i) / \widehat{\text{Var}}(Z_i)\right) \cdot \widehat{\text{Var}}(Z_i)} \\
&= \frac{\widehat{\text{Cov}}(Z_i, Y_i)}{\widehat{\text{Cov}}(X_i, Z_i)} = \hat{\beta}^{IV}
\end{aligned}$$

Implementation in Stata

Instrumental variables is easy to implement in Stata. You should use Stata's built-in command, which will also give you correct standard errors, and can be followed up with additional built-in tests (more on that later). Suppose we want to run the IV regression

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \delta_1 W_1 + \delta_2 W_2 + \varepsilon_i$$

where X_1 and X_2 are the endogenous regressors, W_1 and W_2 are the exogenous regressors, and Z_1 and Z_2 are the excluded instruments. To implement this in Stata, we run:

```
ivregress 2sls y (x1 x2 = z1 z2) w1 w2, rob
```

Note that, as always, we need to specify the option `, rob` to ensure that Stata uses robust standard errors.

Overidentification Test

In general, the orthogonality condition is not testable. In the just-identified case, 2SLS will simply select parameters that set $\widehat{\text{Cov}}(Z_i, \varepsilon_i) = 0$, and so there is no way to independently test that assumption.

However, when we have more instruments than endogenous regressors (when we are overidentified), we can run something called an overidentification test. Since 2SLS will not be able to set $\widehat{\text{Cov}}\left(Z_{j,i}, \varepsilon_i\right) = 0$ for all j in this case, we can test the null hypothesis that all the orthogonality conditions do in fact hold in the population. To do this in Stata, we first run overidentified two-stage least squares, and then we run:

```
estat overid
```

This command will output a test statistic and p-value for the test. If the p-value is below our significance threshold, then we can reject the null that all the orthogonality conditions hold.

A few notes on the overidentification test:

- Intuitively, the overidentification test is a test of whether each instrument would give you the same IV coefficients. If you have two instruments for education and they give very different estimates for the returns to education, that would cause the overidentification test to reject.
- Just because the overidentification test fails to reject does not mean that your instruments are valid. One possibility, of course, is that the test failed to reject because there was not enough statistical power, but would have rejected in a larger sample. But another possibility is that your instruments are wrong, but they are wrong in the same way and so the overidentification test cannot detect it.
- The overidentification test tells you if your instruments are invalid (really, if they do not agree with each other), but it does not tell you how bad the problem is. For even a very small violation of orthogonality, the overidentification test will reject in a sufficiently large sample.
- In modern empirical work, the overidentification test has fallen out of fashion. One reason for this is that different instruments may be picking up causal effects for different groups: this would cause the overidentification test to reject, even if each instrument is identifying a true causal effect (we will talk about this in a few lectures).